Diel thermal variability structures algal and microbial symbiont communities in the reef-building coral, *Siderastrea siderea*

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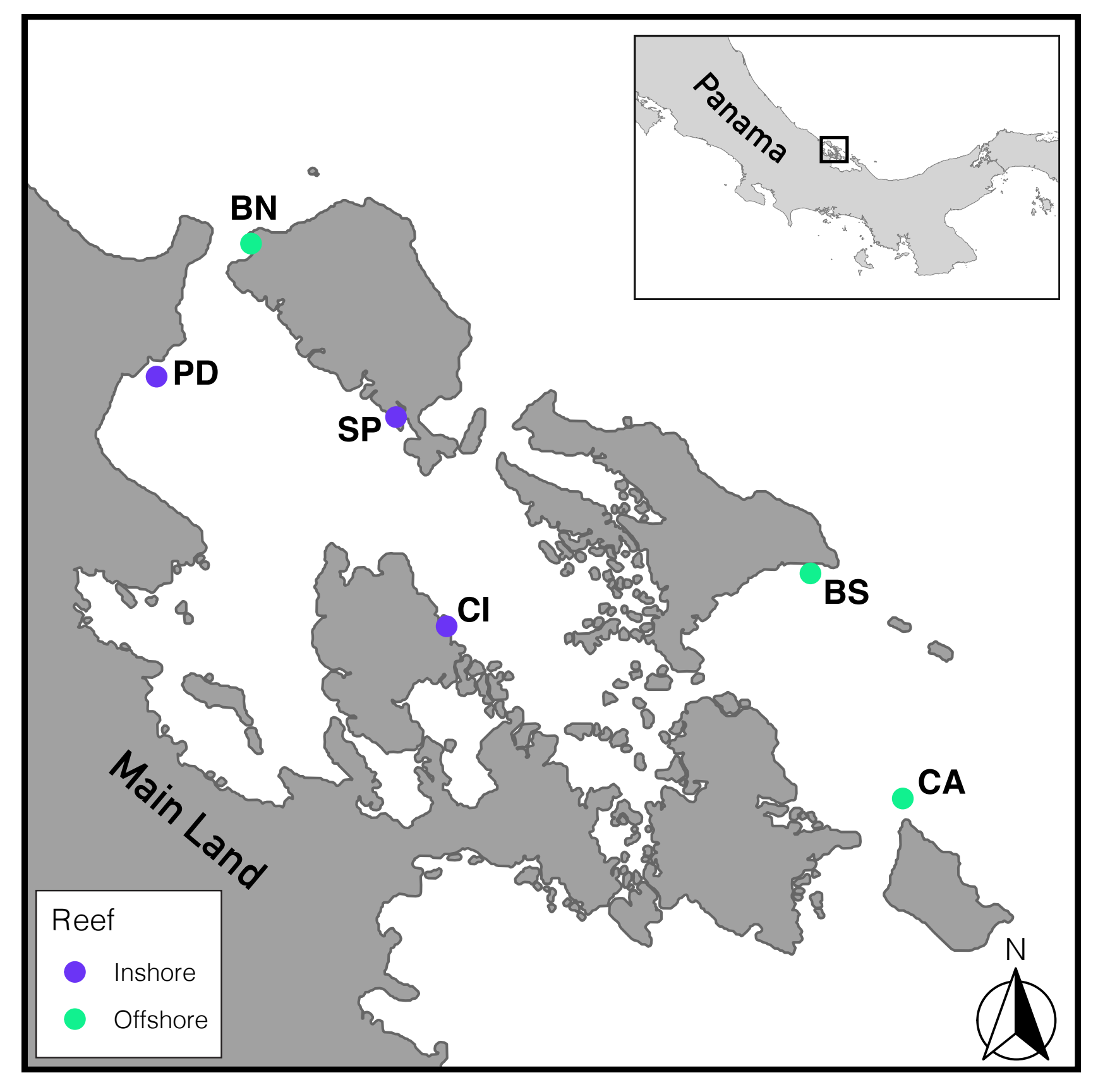
Introduction

- Diel thermal variability (DTV) may play a role in mitigating the effects of coral bleaching (Safaie et al. 2018)
- Association with certain algal and microbial symbionts increases bleaching resistance and and facilitates coral acclimatization (van Oppen et al. 2018; Howells et al 2012)

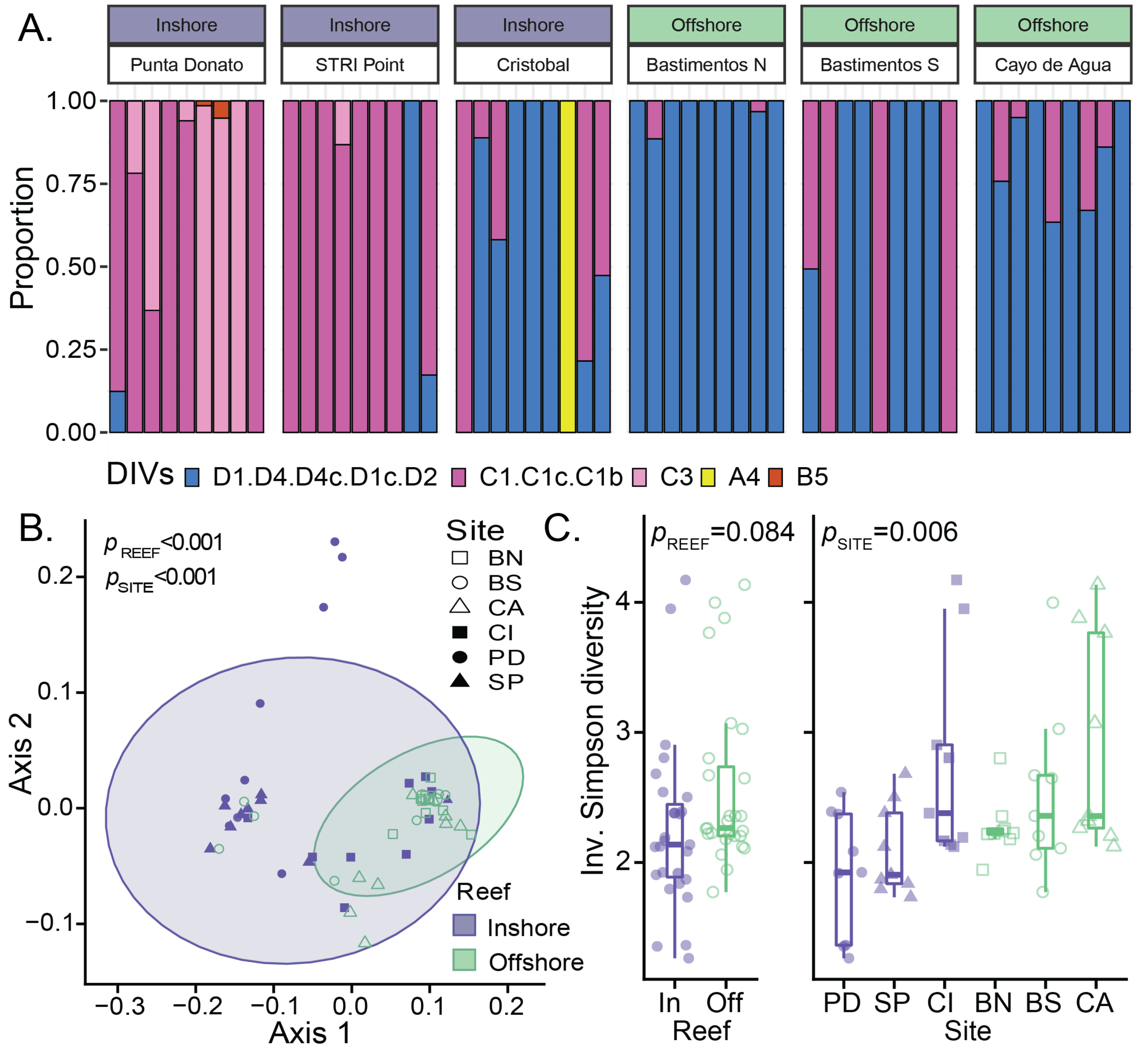
How do different patterns of diel thermal variability influence structure of symbiotic communities of *Siderastrea siderea*?



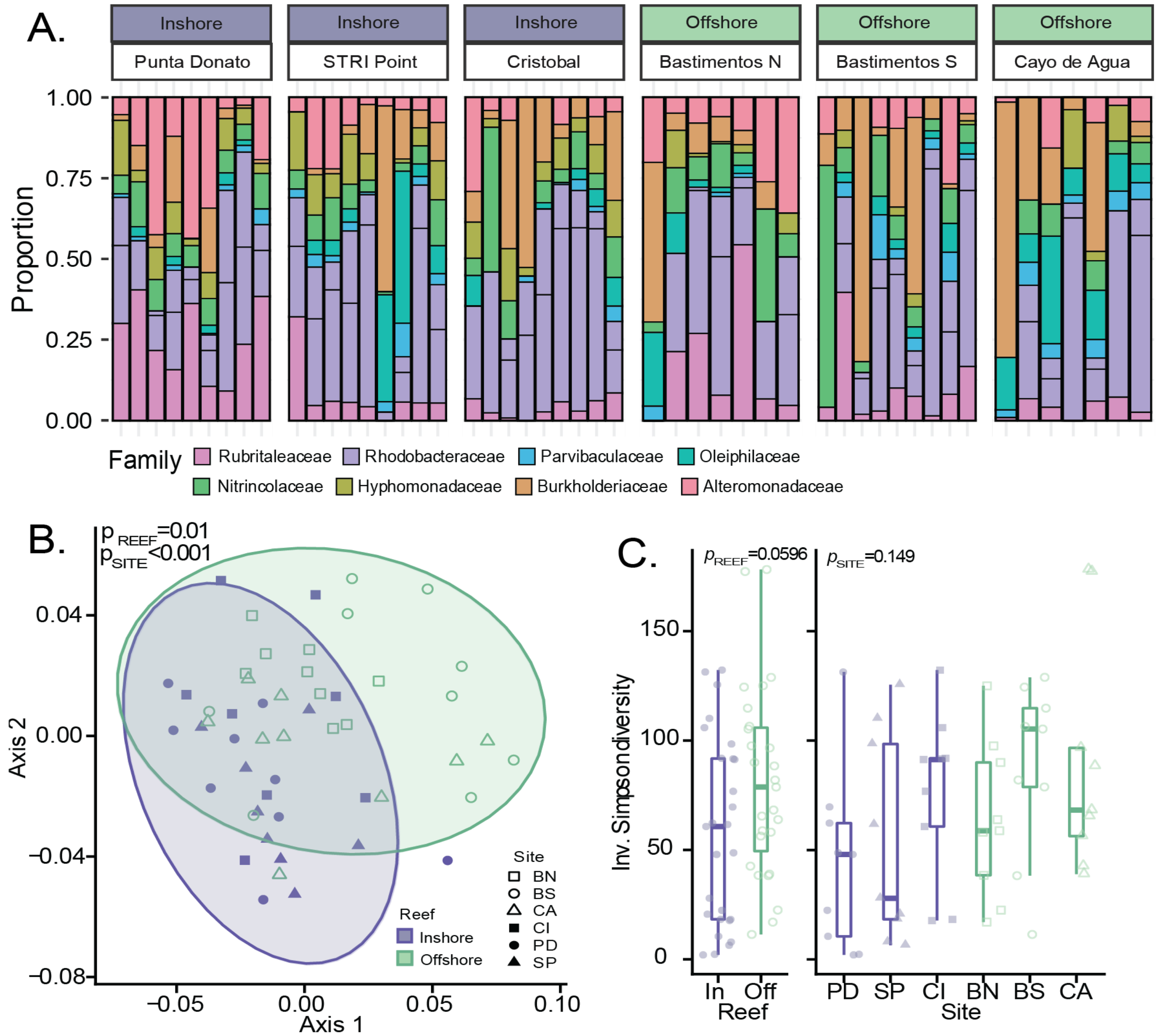
Coral Sampling

- n=54
- 9 colonies/site
- 3 inshore sites
- 3 offshore sites

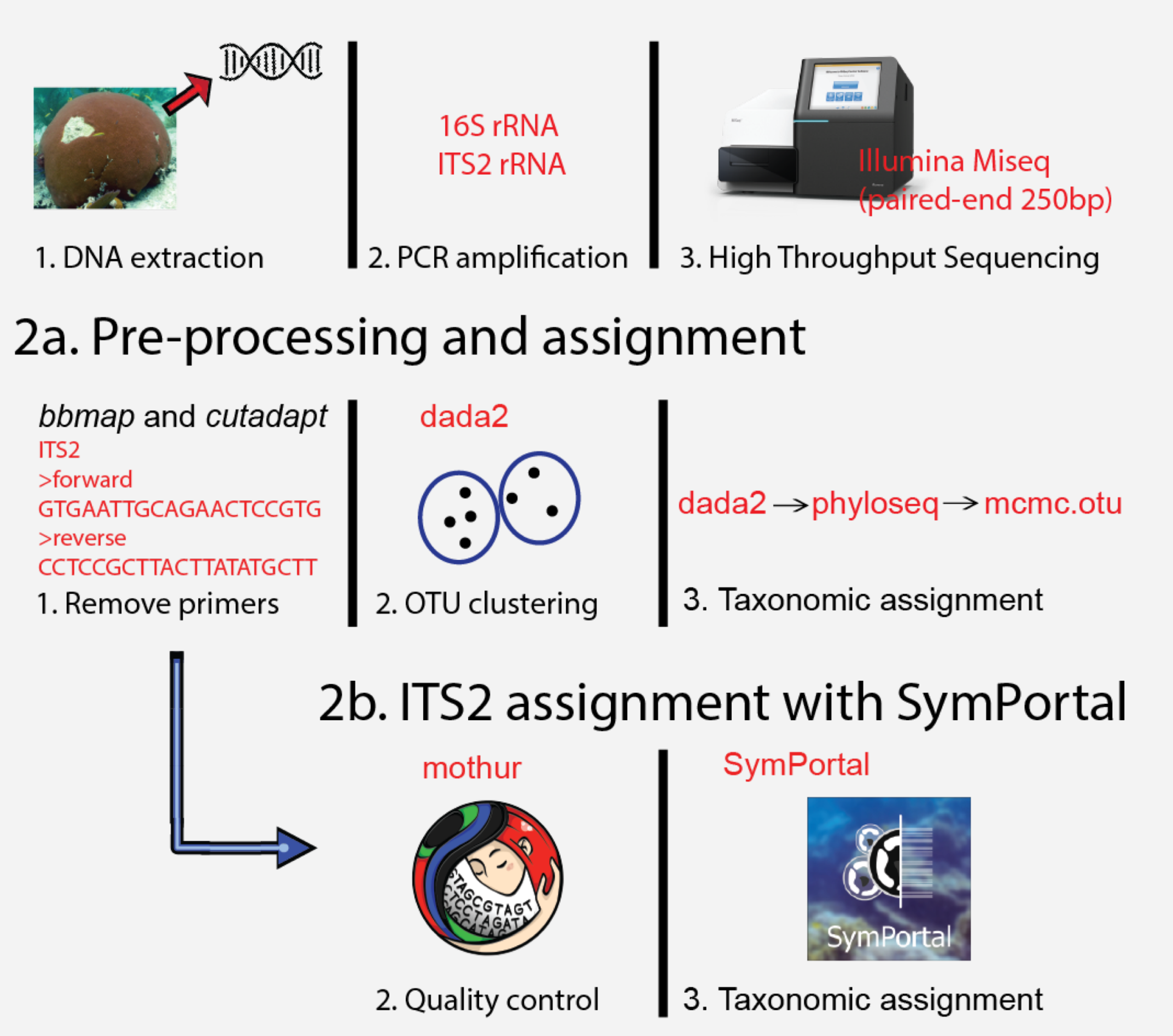
Symbiodiniaceae Characterization



Microbial Characterization



Methods



1. Library preparation

1. DNA extraction

2. PCR amplification
16S rRNA
ITS2 rRNA

3. High Throughput Sequencing
Illumina Miseq
(paired-end 250bp)

2a. Pre-processing and assignment

bbmap and *cutadapt*
ITS2
>forward
GTGAATTGCAGAACTCCGTG
>reverse
CCTCCGCTTACTTATATGCTT
1. Remove primers

dada2
2. OTU clustering

dada2 → phyloseq → mcmc.otu
3. Taxonomic assignment

2b. ITS2 assignment with SymPortal

mothur
2. Quality control

SymPortal
3. Taxonomic assignment

Future Directions

- Continue analysis of microbial community by identifying indicator species and accessory microbiome compared to the core microbiome
- Further research is currently investigating the modulation of algal and microbial communities under experimental conditions

Main Findings

- *Durusdinium* more thermally tolerant than *Cladocopium* C1 (*S. goreaui*) (Oliver and Palumbi 2011); however, we found a greater relative abundance of *Durusdinium* in the cooler offshore reef compared to a greater abundance of C1 in the warmer inshore reef
- *Alphaproteobacteria Hyphomonadaceae* may also be indicative of coral stress; found to increase after bleaching event (Pootakham et al. 2018)
- The diel thermal variability differences that inshore and offshore reefs experience correspond with differing algal and microbial community composition which may indicate that DTV plays a role in structuring these differences

Acknowledgements

- Thank you to Professor Sarah Davies for her knowledge and guidance, Nicola Kriefall and Hannah Aichelman for technical assistance, Karl D Castillo and the Smithsonian Tropical Research Institute (STRI) and permitting agencies, and the BU Marine Program for the opportunity to engage in amazing research experiences

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